

GURMAN AVTAR

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Education

Indian Statistical Institute, Kolkata

Bachelor of Statistical Data Science

Percentage - **83.2%/100%**

Jan 2024 – Present

Work Experience

IIT Kanpur

Summer Intern (SURGE Program)

Onsite

- Engineered a machine learning pipeline utilizing 1D-CNN and tree-based models to predict landslide susceptibility across **9,400+ spatial data points**.
- Developed a custom Auto-VIF algorithm to eliminate multicollinearity and resolved a critical “NaN-bleed” issue, restoring data integrity for **11+ million raster pixels**.
- Implemented Spatial Block Cross-Validation to mitigate spatial data leakage, yielding a robust **89% ROC-AUC**.

Projects

E-commerce Warehouse Location Optimization

- Formulated a Mixed-Integer Linear Program (MILP) in Python (PuLP) optimizing a 4-warehouse configuration based on Poisson-modeled Indian Census demand distributions.
- Projected a **12% reduction** in operational overhead by minimizing aggregate delivery distance and latency across major logistics hubs.

DIN-SQL: Decomposed In-Context Learning of Text-to-SQL

- Developed a few-shot prompting framework decomposing text-to-SQL into sub-tasks: schema linking, query classification, SQL generation, and self-correction.
- Utilized LLMs to process sub-problem solutions, improving simple few-shot baseline performance by **~10%**.
- Achieved state-of-the-art **85.3% execution accuracy** on the Spider dataset and **55.9%** on the BIRD benchmark, outperforming heavily fine-tuned models.

Analytical & Numerical Solutions to Black-Scholes Equation

- Formulated an analytical option pricing model by transforming the Black-Scholes PDE into the 1D Heat Equation via spatial Fourier Transforms.
- Engineered a vectorized Monte Carlo simulation in Python (NumPy) to validate closed-form solutions, proving asymptotic empirical convergence.

Quantitative Trading Strategy

- Developed a weekly rebalancing quantitative trading strategy on NIFTY50 equities using Python.
- Engineered a Voting Classifier ensemble combining Random Forest and HistGradientBoosting to maximize signal accuracy.
- Achieved a **51.3% cumulative return** over a 3-year period with a **54.8% win rate**.

Technical Skills

Python, R, C++, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, PuLP, Matplotlib, ggplot2, Machine Learning, Deep Learning, Statistical Inference, Optimization, Quantitative Modeling, Regression Analysis, Git, GitHub, SQL, LaTeX, VS Code, PyCharm, MS Excel

Achievements

Highest Academic Achiever Award: Merit-based cash prize winner at ISI Kolkata [2025]

IMC Prosperity 3 Trading Competition: **Global Rank 249** (16,000+ teams), **All India Rank 43** [2025]

SPARQ Quant Competition: **Top 20 Rank** in Statistical and Probabilistic Analysis [2026]

ISI Admission Test: Secured **All India Rank 70** [2024]

Certifications

Winter School on Deep Learning

ISI Kolkata

SPARQ Industry Interaction Workshop

Kolkata

Interests

Competitive Programming, Chess, Leadership, Sports & Hobbies